

PENSACOLA BUOY 42039, USA

WMO: 995470

Lat: 28.788N

Lon: 86.008W

Elev: 0

StdP: 101.33

Time Zone: -6.00 (W06)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	9.7	0.4	3.9	10.5	1.6	4.2	10.9	15.3	14.7	14.5	14.0	8.4	350	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.1	30.5	26.3	30.2	26.1	29.9	26.0	27.4	29.4	26.9	29.1	26.7	29.0	3.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.4	28.9	26.3	21.7	28.7	25.8	21.1	28.5	86.3	29.4	84.6	29.2	83.1	29.1	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.8	11.7	10.8	DB	5.3	30.9	2.0	0.5	3.9	31.3	2.7	31.6	1.6	32.0	0.2	32.3	(4)
			WB	3.0	28.1	1.6	0.9	1.9	28.7	0.9	29.2	0.0	29.7	-1.2	30.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	16.7	17.3	19.0	21.6	24.4	27.3	28.1	28.5	27.4	24.8	21.0	18.8	(6)
		DBStd	4.82	3.71	3.23	2.71	2.12	1.60	1.18	0.98	1.09	1.09	2.22	2.97	3.38	(7)
		HDD10.0	3	3	1	0	0	0	0	0	0	0	0	0	0	(8)
		HDD18.3	203	73	51	26	4	0	0	0	0	1	12	37		(9)
		CDD10.0	4727	212	206	278	347	447	519	563	573	523	458	331	272	(10)
		CDD18.3	1886	23	23	46	101	189	269	304	314	273	200	93	51	(11)
		CDH23.3	16640	1	6	11	143	1040	2916	3734	4009	3066	1403	249	61	(12)
(13)	Wind	CDH26.7	4492	0	0	0	0	25	677	1293	1560	811	126	1	0	(13)
		WSAvg	5.7	7.0	6.4	6.2	5.6	5.0	4.1	4.2	4.3	5.6	6.1	6.9	7.1	(14)
		PrecAvg	1363	110	102	127	106	67	137	146	166	139	97	79	88	(15)
		PrecMax	1809	292	264	271	234	214	391	306	316	322	248	208	221	(16)
		PrecMin	888	10	14	6	10	5	37	66	66	43	14	18	33	(17)
		PrecStd	232	64	56	71	59	44	86	51	60	71	57	48	46	(18)
		DB	23.3	24.3	24.5	25.8	28.0	30.2	30.7	31.4	30.1	29.1	26.8	26.0		(19)
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	22.4	22.3	23.0	23.5	24.7	26.0	26.7	26.4	25.4	25.7	24.0	23.8		(20)
		DB	22.5	23.2	23.5	25.3	27.4	29.7	30.2	30.7	29.6	28.6	26.2	25.0		(21)
		MCWB	21.5	21.7	22.2	23.3	24.6	26.2	26.3	26.4	25.2	25.5	24.2	23.4		(22)
		DB	21.9	22.2	22.9	24.8	26.9	29.3	29.9	30.3	29.2	28.0	25.5	24.0		(23)
		MCWB	20.8	21.3	21.6	22.7	24.2	25.9	26.1	26.2	25.2	24.9	23.1	22.7		(24)
		DB	21.5	21.4	22.3	24.3	26.5	29.0	29.6	30.1	29.0	27.4	24.8	23.1		(25)
		MCWB	20.2	20.5	21.0	22.0	23.9	25.7	25.8	26.1	25.1	24.5	22.1	21.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	22.8	22.8	23.8	25.0	26.4	28.0	28.0	28.0	27.3	26.5	25.4	24.2		(27)
		MCDB	23.1	23.6	24.0	25.3	27.1	29.3	29.8	30.0	28.6	27.8	26.1	25.0		(28)
		WB	21.8	22.1	22.8	24.2	25.4	27.2	27.1	27.3	26.8	26.1	24.8	23.9		(29)
		MCDB	22.2	22.8	23.0	24.7	26.0	28.9	29.4	29.6	28.5	27.6	25.9	24.7		(30)
		WB	21.2	21.6	22.1	23.5	25.0	26.6	26.8	26.9	26.3	25.7	23.9	23.3		(31)
		MCDB	21.8	22.2	22.4	24.1	25.8	28.6	29.2	29.4	28.3	27.4	25.3	24.2		(32)
		WB	20.4	20.7	21.3	22.8	24.6	26.2	26.4	26.7	25.9	25.2	22.7	22.1		(33)
(34)	Mean Daily Temperature Range	MCDB	21.3	21.4	21.9	23.5	25.6	28.4	29.1	29.3	28.3	27.1	24.5	23.1		(34)
		MDBR	3.5	3.2	2.6	2.1	1.6	1.7	2.0	2.1	1.9	2.0	2.9	3.2		(35)
		5% DB	3.0	2.9	2.5	2.2	1.5	1.8	2.0	2.0	2.0	1.8	2.9	2.9		(36)
		MCDBR	4.2	4.0	3.4	2.9	1.9	1.8	1.9	1.7	1.8	2.0	3.8	3.7		(37)
		5% WB	3.0	2.8	2.9	2.6	1.4	1.7	2.2	2.2	1.9	1.8	3.0	2.8		(38)
		MCWBR	4.4	3.8	3.9	3.3	1.9	2.1	2.4	2.0	2.0	2.4	4.0	3.7		(39)
		taub	0.360	0.369	0.374	0.403	0.429	0.448	0.469	0.464	0.431	0.396	0.368	0.358		(40)
(41)	Clear-Sky Solar Irradiance	taud	2.503	2.489	2.467	2.391	2.351	2.363	2.311	2.336	2.401	2.473	2.498	2.529		(41)
		Ebn at Noon	878	901	915	894	866	844	826	829	848	859	861	862		(42)
		Edh at Noon	92	101	109	121	126	124	131	126	115	101	91	86		(43)
		RadAvg	3.20	3.98	5.20	6.27	6.96	6.51	5.96	5.61	5.13	4.62	3.59	2.92		(44)
(45)	All-Sky Solar Radiation	RadStd	0.29	0.45	0.44	0.52	0.40	0.44	0.38	0.35	0.38	0.32	0.32	0.26		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+142	+94

Nomenclature: See separate page